Ethan Rao

EthanR@silq.tech | 424-309-8496

Research scientist and manager with demonstrated history of developing commercially viable products and valued intellectual property.

Work Experience

DIRECTOR OF RESEARCH AND DEVELOPMENT | SILQ TECHNOLOGIES | JANUARY 2019-PRESENT

- Manages employees in the design and development of new products and new applications of existing products and services of Silq Technologies
- Represents Silq Technologies in the development process of new products in coordination with academic collaborators, suppliers, and contract manufacturers
- New product testing, including comparison testing with competitor products, bench testing, animals testing and clinical evaluation, and including protocol and test report documentation.
- Develop, communicate, and track product development goals of Silq products.

RESEARCH SCIENTIST | SILQ TECHNOLOGIES | DECEMBER 2016-JANUARY 2019

- Continued development of the Silq surface treatment process for biomedical devices
- Participate in the design, fabrication, and validation of equipment for large scale, rapid surface treatment of biomedical devices including the Silq ClearTract Foley Catheter
- Primary contributor to the design and manufacturing of high performance, non-flammable lithium ion batteries based on IP from undergraduate research.
    - US Patent No: US10,629,880 ENERGY PROVIDING DEVICES AND APPLICATIONS THEREOF

UNDERGRADUATE RESEARCHER | KANER LAB, UCLA | JUNE 2015-DECEMBER 2016

- Developed method to cost effectively modify the surface chemistry of polyolefin membrane separators used in lithium-ion batteries. The project received $100,000 grant from UCLA Physical Sciences Entrepreneurship and Innovation Fund for further development.
    - Primary Author Publication: **Rao, E.,** McVerry, B.T., Borenstein, A., Anderson, M., Jordan, R., Kaner, R.B., Roll-to-Roll Functionalization of Polyolefin Separators for High Performance Lithium Ion Batteries. ACS Applied Energy Materials. ACS Appl. Energy Mater., 2018, 1 (7), pp 3292–3300.
- Played a central role in research regarding a method of applying biofouling resistant surface treatments using a photoactivated azide moiety designed for plastics and PDMS.
    - Primary Author Publication: McVerry, B., Polasko, A., **Rao, E.**, Haghniaz, R., Chen, D., He, N., Ramos, P., Hayashi, J., Curson, P., Wu, C.-Y., Bandaru, P., Anderson, M., Bui, B., Sayegh, A., Mahendra, S., Carlo, D. D., Kreydin, E., Khademhosseini, A., Sheikhi, A., Kaner, R. B., A Readily Scalable, Clinically Demonstrated, Antibiofouling Zwitterionic Surface Treatment for Implantable Medical Devices. *Adv. Mater.* 2022, 34, 2200254.
    - Patent No US10729822B2 BIOFOULING RESISTANT COATINGS AND METHODS OF MAKING AND USING THE SAME
- Contributing author in research involving low bio-adhesion surfaces and novel asymmetric membranes
    - Lin, Cheng-Wei, et al. "Direct Grafting of Tetraaniline via Perfluorophenylazide Photochemistry to Create Antifouling, Low Bio-Adhesion Surfaces." Chemical Science, vol. 10, no. 16, 2019, pp. 4445–4457.
    - McVerry, Brian, et al. "Next-Generation Asymmetric Membranes Using Thin-Film Liftoff." *Nano Letters*, vol. 19, no. 8, American Chemical Society (ACS), July 2019, pp. 5036–43.
    - Aguilar, Stephanie, et al. "Enhancing Polyvalent Cation Rejection Using Perfluorophenylazide-Grafted-Copolymer Membrane Coatings." *ACS Applied Materials & Interfaces*, vol. 12, no. 37, American Chemical Society (ACS), Sept. 2020, pp. 42030–40.

WEBTECHNICIAN | UCLA HISTORY DEPARTMENT | APRIL 2014-AUGUST 2016

- Maintained up-to-date class websites through coordination with faculty members and managed online content.

ASSISTANT WEB DEVELOPER | CALCULATORSOUP.COM | JUNE 2013-AUGUST 2014

- Participated in initial development of Calculatorsoup.com designing useful calculators for students

Skills

- Biomedical Product Development, Research Management, Material Science, Surface Chemistry, Biochemistry, Lithium-ion Battery Research
- Programming: Python, SQL, MATLAB, Data Collection/ Analysis, Image Processing, Website Design.

Education

UNIVERSITY OF CALIFORNIA, LOS ANGELES | SEPTEMBER 2013-DECEMBER 2016

- BS in Biochemistry, Dean’s List 2016.
- Grand Challenge Undergraduate Research Scholar, UCLA iGEM Synthetic Biology Team Member
- Guest Speaker for Chem 147: Careers in Chemistry and Biochemistry (Fall 2018, Winter 2020, Winter 2021), UCLA Physical Science's "Welcome to Research" Panel (Winter 2019).